

## SAFETY DATA SHEET

Revision Date 01-April-2024

Revision Number 4

### 1. Identification

**Product Name** Antimony(V) oxide, 30% in H<sub>2</sub>O, colloidal dispersion

**Cat No. :** 40128

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Acute oral toxicity** Category 4

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Harmful if swallowed

**Precautionary Statements****Prevention**

Do not eat, drink or smoke when using this product  
Wash face, hands and any exposed skin thoroughly after handling

**Response**

IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell  
Rinse mouth

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Harmful to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                                            | CAS-No     | Weight % |
|------------------------------------------------------|------------|----------|
| Water                                                | 7732-18-5  | 64.00    |
| Antimony oxide (Sb <sub>2</sub> O <sub>5</sub> )     | 1314-60-9  | 22.00    |
| Antimonate (SbO <sub>3</sub> <sup>1-</sup> ), sodium | 15432-85-6 | 12.00    |
| Triethanolamine                                      | 102-71-6   | 2.00     |

### 4. First-aid measures

|                                        |                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                          |
|---------------------------------------|--------------------------|
| <b>Suitable Extinguishing Media</b>   | Not combustible.         |
| <b>Unsuitable Extinguishing Media</b> | No information available |
| <b>Flash Point</b>                    | No information available |
| <b>Method -</b>                       | No information available |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Sodium oxides. Antimony oxide.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
-

**6. Accidental release measures**

|                                  |                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                |
| <b>Environmental Precautions</b> | Should not be released into the environment. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. |

|                                             |                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|

**7. Handling and storage**

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Acids. Reducing Agent.                                                             |

**8. Exposure controls / personal protection****Exposure Guidelines**

| Component                                            | Alberta                    | British Columbia           | Ontario TWAEV                              | Quebec                     | ACGIH TLV                  | OSHA PEL                             | NIOSH                                                    |
|------------------------------------------------------|----------------------------|----------------------------|--------------------------------------------|----------------------------|----------------------------|--------------------------------------|----------------------------------------------------------|
| Antimony oxide (Sb <sub>2</sub> O <sub>5</sub> )     | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup>                 | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | (Vacated) TWA: 0.5 mg/m <sup>3</sup> | IDLH: 50 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> |
| Antimonate (SbO <sub>3</sub> <sup>1-</sup> ), sodium | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup>                 | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | (Vacated) TWA: 0.5 mg/m <sup>3</sup> | IDLH: 50 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> |
| Triethanolamine                                      | TWA: 5 mg/m <sup>3</sup>   | TWA: 5 mg/m <sup>3</sup>   | TWA: 0.5 ppm<br>TWA: 3.1 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup>   | TWA: 5 mg/m <sup>3</sup>   |                                      |                                                          |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH - NIOSH - National Institute for Occupational Safety and Health

|                             |                                   |
|-----------------------------|-----------------------------------|
| <b>Engineering Measures</b> | None under normal use conditions. |
|-----------------------------|-----------------------------------|

**Personal protective equipment**

|                        |                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Hand Protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                   |

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Natural rubber | See manufacturers recommendations | -               | Splash protection only |
| Nitrile rubber |                                   |                 |                        |
| Neoprene       |                                   |                 |                        |
| PVC            |                                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

#### Environmental exposure controls

Prevent product from entering drains.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                                |
|-----------------------------------------------|--------------------------------|
| <b>Physical State</b>                         | Solid dispersion               |
| <b>Appearance</b>                             | No information available       |
| <b>Odor</b>                                   | No information available       |
| <b>Odor Threshold</b>                         | No information available       |
| <b>pH</b>                                     | No information available       |
| <b>Melting Point/Range</b>                    | No data available              |
| <b>Boiling Point/Range</b>                    | No information available       |
| <b>Flash Point</b>                            | No information available       |
| <b>Evaporation Rate</b>                       | Not applicable                 |
| <b>Flammability (solid,gas)</b>               | No information available       |
| <b>Flammability or explosive limits</b>       |                                |
| <b>Upper</b>                                  | No data available              |
| <b>Lower</b>                                  | No data available              |
| <b>Vapor Pressure</b>                         | <=1100 hPa @ 50 °C             |
| <b>Vapor Density</b>                          | Not applicable                 |
| <b>Specific Gravity</b>                       | No information available       |
| <b>Solubility</b>                             | Dispersible in water           |
| <b>Partition coefficient; n-octanol/water</b> | No data available              |
| <b>Autoignition Temperature</b>               | No information available       |
| <b>Decomposition Temperature</b>              | No information available       |
| <b>Viscosity</b>                              | Not applicable                 |
| <b>Molecular Formula</b>                      | Sb <sub>2</sub> O <sub>5</sub> |
| <b>Molecular Weight</b>                       | 323.50                         |

## 10. Stability and reactivity

|                        |                                            |
|------------------------|--------------------------------------------|
| <b>Reactive Hazard</b> | None known, based on information available |
|------------------------|--------------------------------------------|

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                                   |
| <b>Conditions to Avoid</b>              | Incompatible products.                                            |
| <b>Incompatible Materials</b>           | Acids, Reducing Agent                                             |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NO <sub>x</sub> ), Sodium oxides, Antimony oxide |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                          |
| <b>Hazardous Reactions</b>              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. Category 4.  
ATE = 300 - 2000 mg/kg.

##### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Mist LC50

Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.

#### Component Information

| Component                                           | LD50 Oral                 | LD50 Dermal                                 | LC50 Inhalation             |
|-----------------------------------------------------|---------------------------|---------------------------------------------|-----------------------------|
| Water                                               | -                         | -                                           | -                           |
| Antimony oxide (Sb <sub>2</sub> O <sub>5</sub> )    | Not listed                | Not listed                                  | LC50 > 5.4 mg/L ( Rat ) 4 h |
| Antimonate (SbO <sub>3</sub> <sup>-</sup> ), sodium | Not listed                | Not listed                                  | LC50 > 5.4 mg/L ( Rat ) 4 h |
| Triethanolamine                                     | LD50 = 4190 mg/kg ( Rat ) | >16 mL/kg ( Rat )<br>>2000 mg/kg ( Rabbit ) | Not listed                  |

#### Toxicologically Synergistic Products

No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

##### Irritation

No information available

##### Sensitization

No information available

##### Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                           | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Water                                               | 7732-18-5  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Antimony oxide (Sb <sub>2</sub> O <sub>5</sub> )    | 1314-60-9  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Antimonate (SbO <sub>3</sub> <sup>-</sup> ), sodium | 15432-85-6 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Triethanolamine                                     | 102-71-6   | Not listed | Not listed | Not listed | Not listed | Not listed |

##### Mutagenic Effects

No information available

##### Reproductive Effects

No information available.

##### Developmental Effects

No information available.

##### Teratogenicity

No information available.

##### STOT - single exposure

None known

##### STOT - repeated exposure

None known

##### Aspiration hazard

No information available

##### Symptoms / effects, both acute and delayed

No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component       | Freshwater Algae                                                                                         | Freshwater Fish                                                                                                                                                                                | Microtox                 | Water Flea |
|-----------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------|
| Triethanolamine | EC50: = 169 mg/L, 96h<br>(Desmodesmus subspicatus)<br>EC50: = 216 mg/L, 72h<br>(Desmodesmus subspicatus) | LC50: 10600 - 13000 mg/L,<br>96h flow-through<br>(Pimephales promelas)<br>LC50: > 1000 mg/L, 96h<br>static (Pimephales promelas)<br>LC50: 450 - 1000 mg/L, 96h<br>static (Lepomis macrochirus) | EC50 > 10000 mg/L 30 min | Not listed |

**Persistence and Degradability** Insoluble in water May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component       | log Pow |
|-----------------|---------|
| Triethanolamine | -2.53   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

### International Inventories

| Component                                            | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Water                                                | 7732-18-5  | X   | -    | X    | ACTIVE                                        | 231-791-2 | -      | -   |
| Antimony oxide (Sb <sub>2</sub> O <sub>5</sub> )     | 1314-60-9  | X   | -    | X    | ACTIVE                                        | 215-237-7 | -      | -   |
| Antimonate (SbO <sub>3</sub> <sup>1-</sup> ), sodium | 15432-85-6 | X   | -    | X    | ACTIVE                                        | 239-444-7 | -      | -   |
| Triethanolamine                                      | 102-71-6   | X   | -    | X    | ACTIVE                                        | 203-049-8 | -      | -   |

| Component                                            | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Water                                                | 7732-18-5  | X     | KE-35400 | X    | -    | X    | X    | X     | X     |
| Antimony oxide (Sb <sub>2</sub> O <sub>5</sub> )     | 1314-60-9  | X     | KE-09841 | X    | X    | X    | X    | X     | X     |
| Antimonate (SbO <sub>3</sub> <sup>1-</sup> ), sodium | 15432-85-6 | X     | KE-31355 | X    | X    | X    | X    | X     | X     |
| Triethanolamine                                      | 102-71-6   | X     | KE-25940 | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                   | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Antimony oxide (Sb2O5)      | Part 1, Group A Substance                            |                                                                            |                                           |
| Antimonate (SbO31-), sodium | Part 1, Group A Substance                            |                                                                            |                                           |
| Triethanolamine             | Part 4 Substance                                     |                                                                            |                                           |

## Other International Regulations

### Authorisation/Restrictions according to EU REACH

| Component                   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Antimony oxide (Sb2O5)      | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Antimonate (SbO31-), sodium | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                   | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Water                       | 7732-18-5  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Antimony oxide (Sb2O5)      | 1314-60-9  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Antimonate (SbO31-), sodium | 15432-85-6 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Triethanolamine             | 102-71-6   | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                   | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water                       | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Antimony oxide (Sb2O5)      | 1314-60-9  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y27                      |
| Antimonate (SbO31-), sodium | 15432-85-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y27                      |
| Triethanolamine             | 102-71-6   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

Prepared By

Product Safety Department  
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www.thermofisher.com

**Revision Date** 01-April-2024  
**Print Date** 01-April-2024  
**Revision Summary** New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**